

A Counterexample to Conjecture 00000001034: a Nontrivial Perfect Lee Code in Dimension 2

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Abstract

Conjecture 00000001034 asserts that nontrivial perfect codes in the Lee metric exist only in dimension $n = 1$ (with $q \geq 5$) and $n = 3$ (with $q = 2$). The code $\{(x, y) \in \mathbb{F}_5^2 : x + 2y = 0\}$ is a perfect Lee code of radius 1 with five codewords, in dimension $n = 2$. This is the classical two-dimensional perfect Lee code of Golomb and Welch. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000001034 reads:

Definition: Perfect codes in the Lee metric are ball-packing partitions of $\mathbb{F}_q^n$ by Lee balls. Conjecture: Nontrivial perfect Lee codes exist only for $n = 1$ ($q \geq 5$) and $n = 3$ ($q = 2$, the Golomb–Welch perfect codes); for $n \geq 4$ all Lee perfect codes are trivial; in particular the Golomb–Welch counterexamples are excluded outside $(q, n) = (5, 3)$.

2 Reading of the statement

The Lee metric lives on $\mathbb{Z}/q$; we take q prime, so that $\mathbb{F}_q = \mathbb{Z}/q$, which covers the case used below. The *Lee weight* of $a \in \mathbb{Z}/q$, with representative $0 \leq a < q$, is $\min(a, q - a)$. The *Lee distance* between $x, y \in (\mathbb{Z}/q)^n$ is $\sum_i \text{wt}(x_i - y_i)$.

Definition 1. $C \subseteq \mathbb{F}_q^n$ is a *perfect Lee code of radius e* if the Lee balls of radius e around the codewords partition $\mathbb{F}_q^n$, i.e. every word lies within Lee distance e of exactly one codeword. It is *nontrivial* if $e \geq 1$ and $|C| \geq 2$.

The conditions $e \geq 1$ and $|C| \geq 2$ exclude the two degenerate cases: radius 0 ($C = \mathbb{F}_q^n$), and a single ball covering everything. The first sentence of the conjecture asserts: if a nontrivial perfect Lee code exists in $\mathbb{F}_q^n$, then $n = 1$ and $q \geq 5$, or $n = 3$ and $q = 2$.

3 Result

Theorem 2. $C = \{(0, 0), (1, 2), (2, 4), (3, 1), (4, 3)\} = \{(x, y) \in \mathbb{F}_5^2 : x + 2y = 0\}$ is a perfect Lee code of radius 1 in $\mathbb{F}_5^2$. Since $n = 2$, Conjecture 00000001034 is false.

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4 Proof

A Lee ball of radius 1 in $\mathbb{F}_5^2$ consists of its centre c and the four points $c \pm (1, 0)$, $c \pm (0, 1)$, so it has 5 points. Five balls therefore have $25 = |\mathbb{F}_5^2|$ points in total, and it suffices to show that the balls are pairwise disjoint. By the triangle inequality, this holds if distinct codewords are at Lee distance at least 3.

The code is the kernel of the linear map $\varphi(x, y) = x + 2y$, so the difference v of two distinct codewords is a nonzero vector with $\varphi(v) = 0$. The nonzero vectors of Lee weight at most 2 are $\pm e_1$, $\pm e_2$, $\pm 2e_1$, $\pm 2e_2$ and $\pm e_1 \pm e_2$. Their φ -values are ± 1 , ± 2 , ± 2 , ± 4 , ± 3 and ± 1 , none of them 0 in $\mathbb{F}_5$. Hence distinct codewords are at distance at least 3, and the balls partition $\mathbb{F}_5^2$. The Lean development checks the defining property directly, over all 25 words.

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `leeWeight`, `leeDist` — the Lee weight on `ZMod q` and the Lee distance on `Fin n -> ZMod q`.
- `IsPerfectLeeCode q n e C` — every word x has exactly one codeword within Lee distance e .
- `ConjectureHolds` — for every prime q , every n , $e \geq 1$ and every perfect Lee code C with $|C| \geq 2$: $(n = 1 \wedge q \geq 5) \vee (n = 3 \wedge q = 2)$.
- `C5`, `isPerfectLeeCode_C5`, `card_C5` — the code of Theorem 2, checked by `decide`.
- `conjecture_00000001034_false` — $\neg \text{ConjectureHolds}$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000001034_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Golomb and Welch [1] showed that perfect Lee codes exist in dimensions 1 and 2 for every radius, and for radius 1 in every dimension. Their celebrated conjecture is that there are none for $n \geq 3$ and $e \geq 2$. The list in Conjecture 00000001034 therefore omits dimension 2 entirely. Its claim that $n \geq 4$ admits only trivial codes also fails, already for $q = 2$, where the Lee metric is the Hamming metric and the binary Hamming code of length 7 is perfect of radius 1. We formalize only the dimension-2 counterexample.

References

- [1] S. W. Golomb and L. R. Welch, *Perfect codes in the Lee metric and the packing of polyominoes*, SIAM J. Appl. Math. 18 (1970), 302–317.